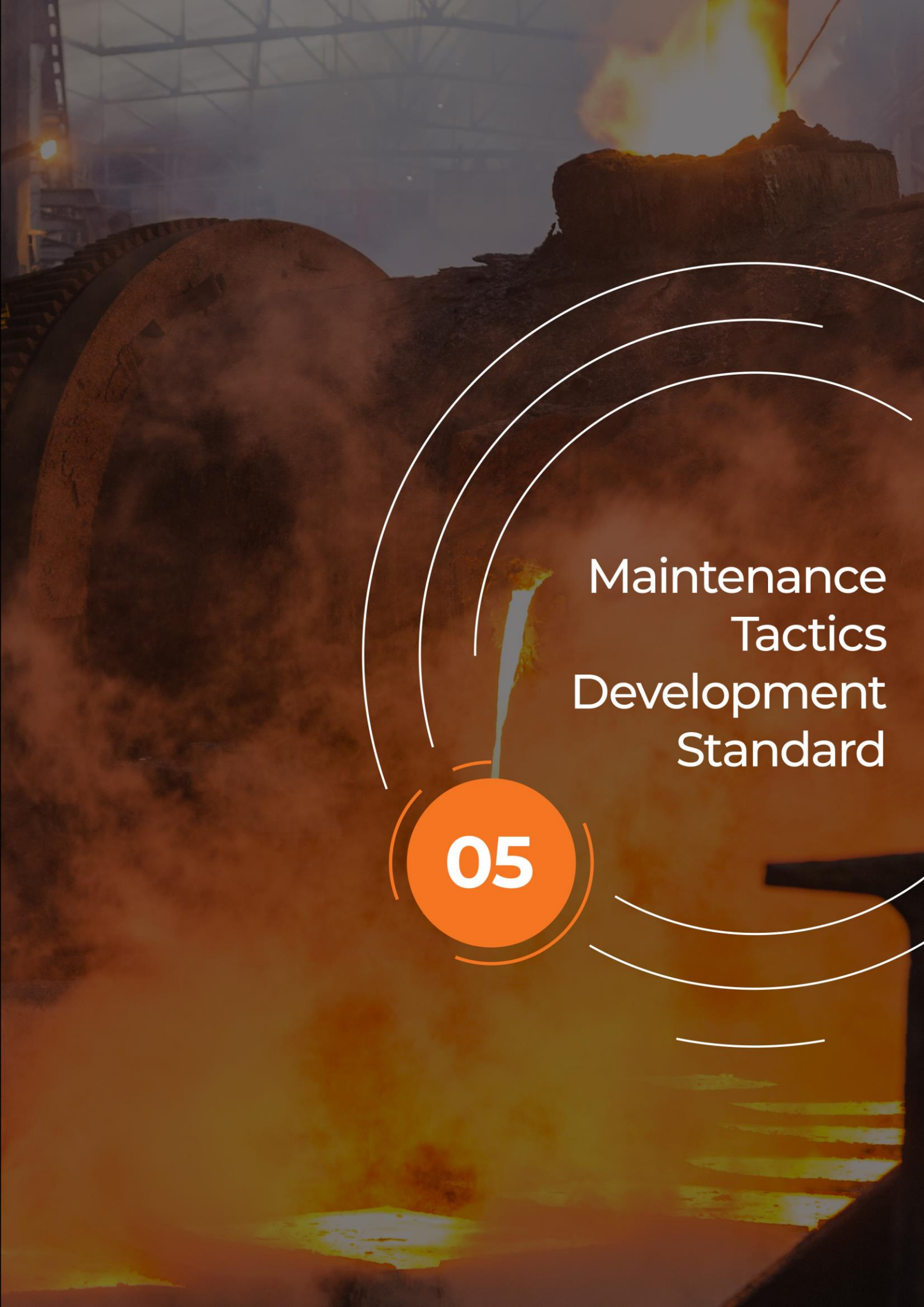


Maintenance Tactics Development Standard

05

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Asset Management, Glencore Copper
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Preface

The Asset Management Framework, Policy, Manuals and Standards are intended to provide leading Asset Management practices when applied and will be continually improved for the benefit of Glencore Copper.

The real value of sharing the knowledge gained from the use of these documents will come from continued discussion of the issues and rationale behind each document rather than by reading and interpreting them as individuals.

Readers are encouraged to consider these against our values around Safety, Responsibility, Simplicity, Integrity, Openness and Entrepreneurialism and how they reflect our Purpose – to responsibly source the commodities that advance everyday life. This document lays out the minimum standard for Glencore Copper and compliance to this procedure will demonstrate a maturity value equal to Planned/Deployed (3) on the Glencore Copper Asset Management Maturity Scale.

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1. Introduction

The Asset Management Policy, Manuals and Standards define how Glencore Copper manages the whole of life of an asset to ensure optimal value is extracted by ensuring that assets are capable of meeting business and stakeholder objectives.

Each standard provides enough detail to implement the minimum requirements at each Glencore Copper region/site. They provide detail below the Asset Management Framework and supporting element Manuals.

Manual E05 Equipment Operating and Maintenance Strategy sets out the Level 2 process requirements of which process steps 2.0 Asset Tactics Developed and 4.0 Asset Tactics Optimised form part of that process. These process steps require the responsible person to develop/update maintenance tactics using one of three (3) methodologies:

- Reliability Centred Maintenance (RCM)
- Failure Mode and Effects Analysis (FMEA)
- Plant Maintenance Optimisation (PMO)

This standard provides the process by which this can be achieved.

The term asset used in this standard is implied to include all operational plant and equipment within the Glencore Copper business.

Figure 1-1 illustrates the Level 2 Process Flow for Element 05.

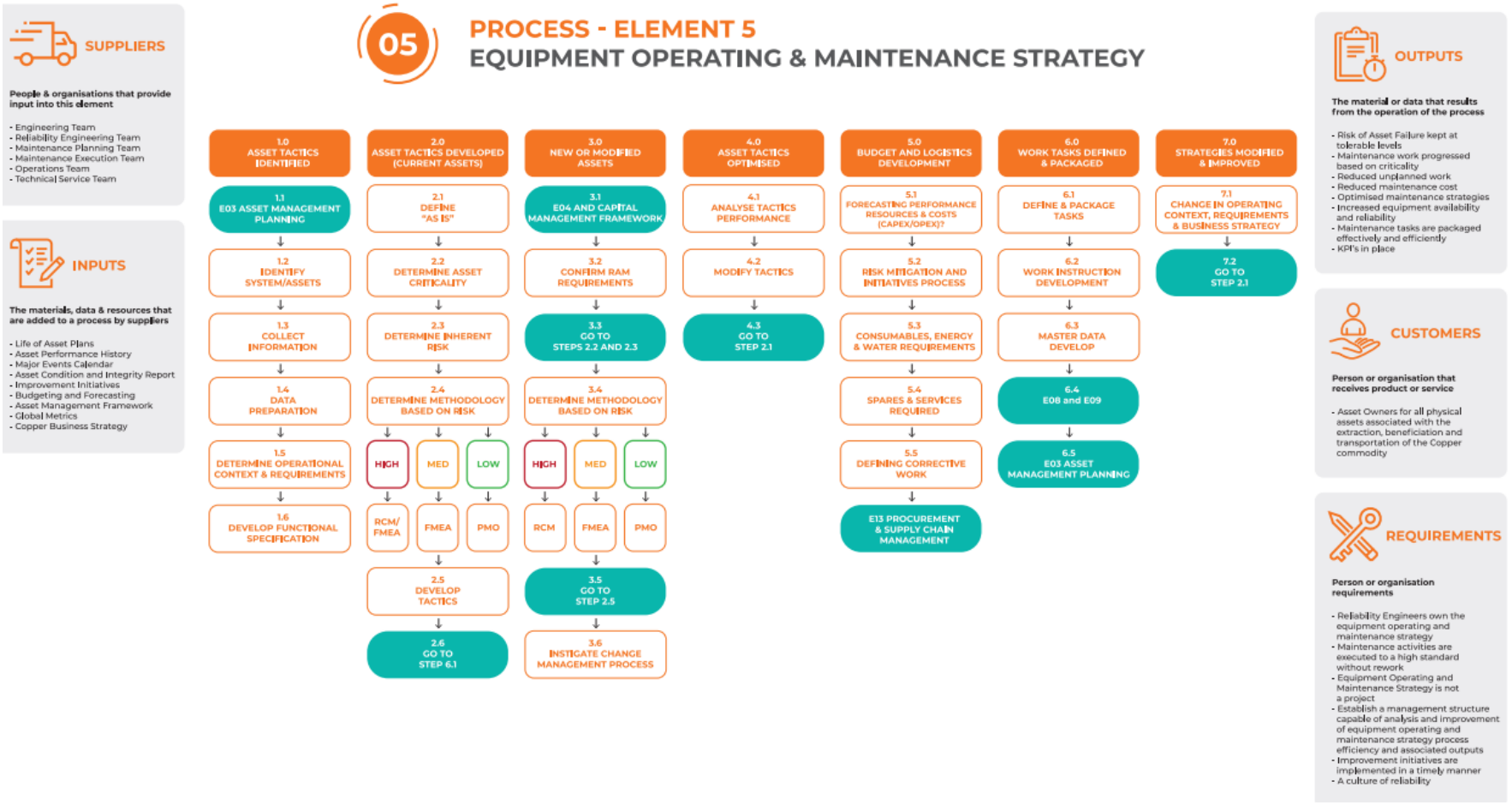


FIGURE 1-1 EQUIPMENT OPERATING AND MAINTENANCE STRATEGY PROCESS FLOW

2. Objectives

The objective of this standard for Maintenance Tactics Development (MTD) is to define the process for developing/modifying and deploying maintenance tactics using one of three methodologies at each Glencore Copper Region/Site. The output for the process is to translate the maintenance tactics into executable maintenance programs.

2.1 Accountabilities

Dedicated resources must be assigned, and capabilities developed within each Operation to identify the key aspects required to ensure that all assets within each Glencore Copper region/operation are identified, operating and maintenance tactics developed or modified, and the tactics entered into or adjusted in the Computerised Maintenance Management System (CMMS).

- Assign accountability for each or all of the following:
  - System and Asset identification.
  - Asset Criticality analysis as required.
  - Operating and maintenance tactics developed for all assets.
  - Operating and maintenance tactics grouped into logical work packages.
  - Tactics authorised.
  - Work instructions developed.
  - Master data loaded into or modified in the CMMS.

2.2 Scope

The scope of this standard is the development/modification and deployment of maintenance tactics by adopting a risk based approached using asset criticality as a means by which the methodology to be used is identified.
Asset criticality is to be determined using the Glencore Copper Asset Criticality Analysis Standard

Reliability Centred Maintenance (RCM) must be applied to all new high (A) critical assets acquired by each Glencore Copper site. For existing assets each site has the option whether RCM is chosen as the methodology for the development of maintenance tactics for high (A) critical assets, but FMEA must be used as a minimum for both high (A) and medium (B) critical assets. The process for both RCM and FMEA is similar, with the difference being that FMEA does not require the assessment of:

- Operating Context
- Functions and Performance Standards
- Functional Failures

The third level of analysis that is to be applied to low (C) critical assets, that being Plant Maintenance Optimisation (PMO).

PMO is a methodology that aims to enhance current maintenance tactics of low (C) critical assets by:

- Optimising tasks frequency and duration (provides cost saving)
- Optimising the number of resources required to complete each task.
- Adopting Original Equipment Manufacturers (OEM) recommended maintenance – saves time over applying RCM or FMEA. Should OEM recommended maintenance not be available it is advised to adopt maintenance tactics for similar assets and adjusted as necessary. Else, new maintenance tactics are to be developed in consultation with a Subject Matter Expert (SME)
- Re-grouping maintenance tasks to make planning easier and execution more effective and efficient.

Note: The PMO process could yield a Operate to Failure (OTF) approach, in which case cost benefit analysis should be conducted to determine if the cost of doing maintenance on these assets is more beneficial than not conducting maintenance. For example:
In the presence of redundancy, a duty asset is allocated an RTF strategy, therefore from a risk and functional perspective it would be an appropriate strategy. However, it may be cost beneficial to conduct maintenance on that asset, particularly if it is an expensive item, as this could yield a cost benefit through deferred replacement.

The maintenance tactics development process described through Reliability-Centred Maintenance is in conformance with the Society of Automotive Engineers (SAE) Standard SAE JA 1011: Evaluation Criteria for Reliability-Centred Maintenance Processes.

This standard is a constituent part of the Equipment Operating and Maintenance Strategy Manual, which in turn is an integral part of the Glencore Copper Asset Management Framework.

MTD is a facilitated process, which is led by an experienced and qualified facilitator. From a purist perspective the tradition method is both time consuming and costly to have every step of the process undertaken through a facilitated team environment. Consideration should be given to the development of asset category failure mode models, which can be used as the basis for developing maintenance tactics through RCM and FMEA.

3. Definitions

Asset Criticality - is the value a business assigns to its assets based on the impact to the business when the asset functional failure occurs.

Asset Criticality Analysis – a process that enables the ranking of systems assets in order of business importance to ensure that effort applied is directionally proportionate to the criticality and impact to the business. Criticality is relevant for the business decision-making process to manage business risk and value. Criticality is typically used to determine:

- As an input to determine the overall priority for performing a maintenance task (sometimes combined with a "Work Order Priority" entered against the specific task to give an overall priority for the task)
- To determine, at a high level, the type of risk mitigation strategy to be applied to the asset (e.g., do condition monitoring and defect elimination on high criticality items)
- As an input into determining the optimum spare parts holding and priority of spares purchasing required for the asset.
- To provide input into the capital program so that "high criticality" asset is given a higher priority for upgrade or replacement.
- To guide reliability engineers so that they focus their reliability improvement efforts on the most "critical" asset.

Availability - Percentage of Calendar Time an asset or plant was physically available to be used.

Critical Asset - asset having potential to significantly impact on the achievement of the organisation's objectives".

Critical System – a productive system comprising of maintainable assets (critical asset) that if unavailable would have a significant risk, cost, and production impact to Glencore Copper.

Evident Failure – a failure mode that become evident to the operating crew under normal operating conditions.

Failure Consequence - the way that a functional failure matters, either directly or indirectly, with regard to safety, environmental, operational capability, or direct and indirect repair costs.

Failure Effect – what happens when a failure occurs.

Failure Mode – a single event that causes functional failure.

FMEA – Failure Mode and Effect Analysis, is a structured approach to discovering potential failures that may exist within a product, asset, or process. Failure modes are the ways in which a process can fail. Effects are the ways that these failures can lead to waste, defects, or harm. Failure Mode and Effects Analysis is designed to identify, prioritize, and limit these failure modes.

Function – what an asset is required to do in terms of defined and measurable performance capability, both primary and secondary functions.

Functional Failure – a state in which an assets or system is unable to perform a specific function to the desired level of performance.

Hidden Failure – a failure mode that is not evident to the operating crew under normal operating condition.

Maintainable Function – an asset that produces part of the overall output. E.g., haul truck engine. Typically, a description from a hierarchical plant register, an assembly of parts, which perform a significant function as a stand-alone item

Maintainable Unit– a group of assets that produces an output from an input. e.g., a haul truck, a pump (pump, motor, coupling, instruments etc.) Typically, a description from a hierarchical plant register representing the Productive Unit, i.e., the parent of the Maintainable Item. A grouping of Maintainable Items which perform a series of key functions.

Mean Time Between Failure - quantifies asset Reliability and represents the average Utilised Time (UT) between Unscheduled Maintenance (UM) events

Multiple Failure - an event that occurs if a protected function fails while its protective device or protective system is in a failed state.

On-condition Task – a maintenance task used to detect potential failures.

Operating Context – The current condition, environment, and culture in which you operate assets.

Primary Function - the intended purpose of an asset that gives reason for its existence, defined in terms of required performance capabilities. An asset may have more than one primary function.

P-F Interval - The interval between the point at which a potential failure becomes detectable and the point at which it causes a functional failure.

Reliability – is the probability that a device will function as designed, within its operating context, for a specified period of time.

Reliability Centred Maintenance - provides a structured framework for analysing the functions and potential failures for a physical asset with a focus on preserving system functions, rather than preserving assets. RCM is used to develop scheduled maintenance plans that will provide an acceptable level of operability, with an acceptable level of risk, in an efficient and cost-effective manner.

Scheduled Discard - a task that entails discarding an item or component at or before a specified age limit regardless of its condition at the time.

Secondary Function - supplementary functions, which an asset is, required to perform in addition to its primary functions. Secondary functions will typically cover environmental controls, safety features, structural features, control systems, containment, prevention of contamination, appearance, protective devices, economy, and efficiency.

Scheduled Restoration - a task that restores the initial capability of an item or component at or before a specified age limit, regardless of its condition at the time.

Sub Process - The means by which resources are transformed from inputs to create useful goods and services as outputs.

4. Approach

The approach for Maintenance Tactics Development (MTD) is described through:

- Process Flow - pictorially illustrates the key MTD activities, refer to Figure 4-1.
- Process Description – defines in more detail the key MTD activities, refer to Section 4.2.
- Process RACI Matrix - indicates for each activity, the functions who are Responsible, Accountable, who are Consulted prior to final decision/action, and those who are Informed of any outcomes and actions, refer to Section 4.3

Maintenance Tactics Development Standard

4.1 Process Flow

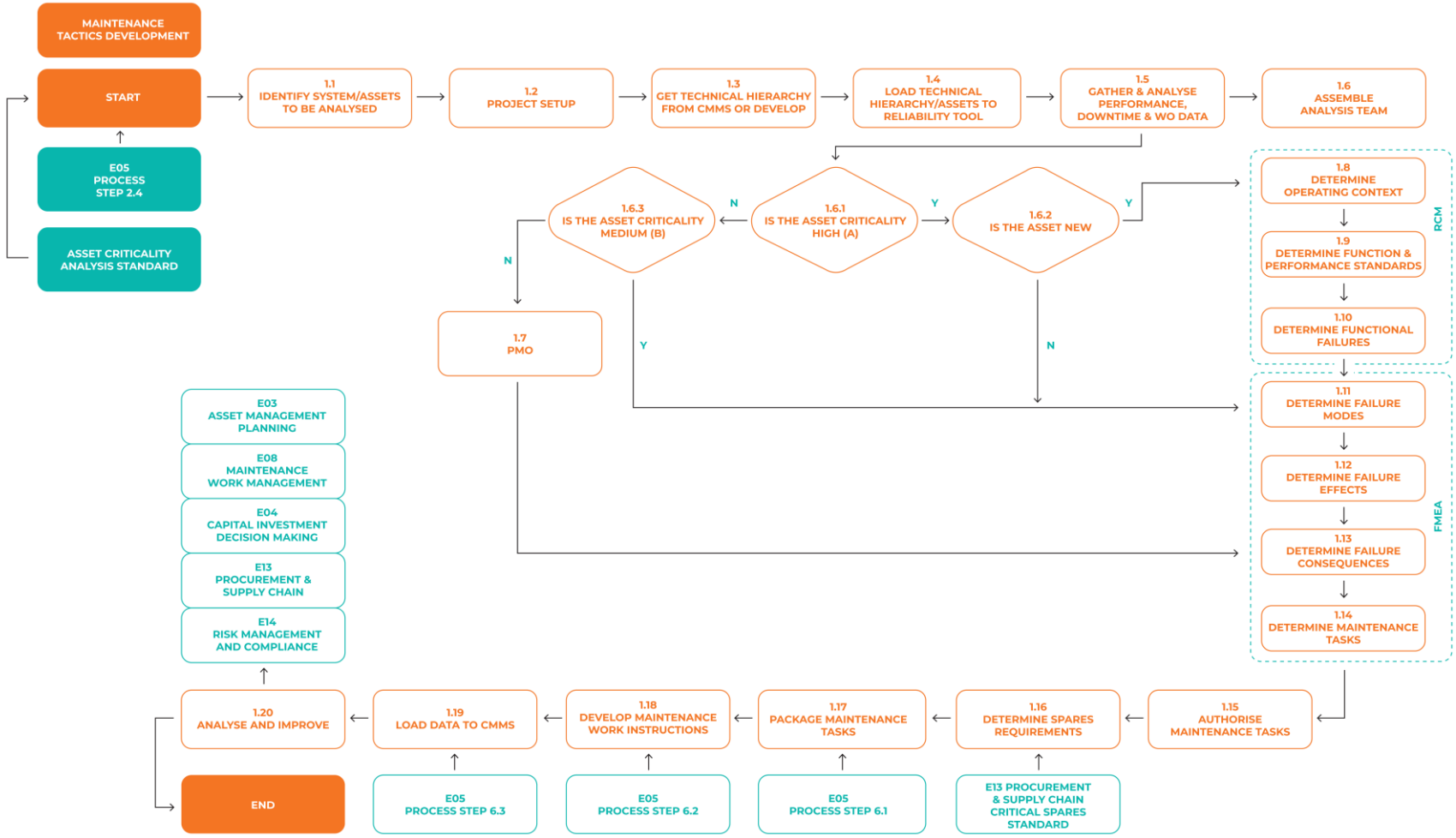


FIGURE 4-1 MAINTENANCE TACTICS DEVELOPMENT PROCESS FLOW

4.2 Process Description

TABLE 4-1 PROCESS DESCRIPTION

PROCESS STEP	DESCRIPTION
1.1 Identify Systems/Assets to be Analysed	Refer to Glencore Copper Criticality Analysis procedure and region/site Critical Asset register and Conduct Criticality analysis as required. This process step is conducted by the system/asset SME and not in a facilitated workshop. Critical Systems/Assets shall be identified that: • Have poor performance. • Have significant cost increase. • Have assets at end of life. • Have assets that are obsolete. • Have not had maintenance tactics developed. • New systems/assets added to the site. • Assessed as due for replacement due to technology changes and benefits.
1.2 Project Set Up	The following activities should be considered: • Business Case Requirements – conduct cost benefit analysis for to justify MTD effort. • Project Charter and Plan Development • Systems and Tool requirements – is software required to conduct MTD and to implement the outcome? • Resource requirements – who will form part of the RCM team and leader to be nominated? • Information System and Technology requirements • External assistance – are external consultants/contractors required to augment the MTD team. • Project office location and set up.
1.3 Get Technical Hierarchy from CMMS or Develop	Download the Technical Hierarchy from the CMMS for those systems to be analysed. If the asset does not have a criticality assigned complete asset criticality analysis as per Glencore Copper Asset Criticality Analysis Standard. If the system/asset does not exist, it is to be created in line with Glencore Copper Master Data Standards.
1.4 Load Technical Hierarchy/Items into Reliability Tool	The technical hierarchy/items to be loaded into the Reliability tool or equivalent.
1.5 Gather and Analyse Performance, Downtime and Work Order Data	Other data to be gathered and analysed. This could include the following: • System/Asset Failure History • System Asset Historical Cost • System/Asset performance criteria (Downtime, MTBF, Availability, Reliability Etc.) • Work order history

1.6 Assemble Analysis Team	A typical Analysis team would consist of members from multiple functions within the business who have intimate knowledge of the systems and assets under review. Such as: • Maintenance – supervisors and trades • Operations – supervisors and trades • Engineering – reliability, mechanical, electrical and instrumentation • Technical Services – process engineers • Finance (as required) A Facilitator will lead the MTD process and should be an experienced Reliability Centred Maintenance (RCM) practitioner and would generally be a Reliability Engineer. The facilitator will train the MTD team, in RCM, FMEA and PMO fundamentals.
1.6.1 Is the Asset Criticality High (A)	If the asset is of High Criticality (A) rank, then go to process step 1.6.2 if not then step 1.6.3
1.6.2 Is the Asset New	If the asset that is under analysis is new and it is High Criticality (A) then the full RCM process is to be followed, therefore go to process step 1.8 and Determine Operating Context. If it is not new but still High Criticality (A) then FMEA part of the process is to be followed, therefore go to process step 1.11 and Determine Failure Modes. Alternatively, each site may want to consider doing RCM on some or even all of the critical assets, but this standard does not prescribe this at this time.
1.6.3 Is the Asset Criticality Medium (B)	If yes go to process step 1.11 and Determine Failure Modes, and conduct FMEA part of the process, this is dictated in process step 2.4 in Element E05 Equipment Operating and Maintenance Strategy of the Asset Management Framework Manual. If no, then PMO is to be conducted and the consequences of failure considered determining if maintenance tasks are worth doing from a purely financial perspective. Therefore, go to process step 1.13 and Determine Failure Consequences. For example, conducting vibration analysis on rotating assets provides sufficient failure data and predictability of the onset of functional failure, which is often considered more cost effective to adopt this strategy, as opposed to a operate to failure strategy. This is due to the potential increase in asset life and increased time between replacement/repair. Therefore, the cost of doing maintenance should be compared to the cost of doing no maintenance for the same analysis period, to determine the best cost option.
1.7 PMO	Maintenance tactics for low criticality (C) assets shall be challenged in the first instance for validity as described in process step 1.6.3. As described in Section 2.2 of this standard, PMO is a methodology that aims to enhance current maintenance tactics by: • Optimising tasks frequency and duration – challenge how often tasks are being scheduled and executed and analysis to see if the frequency of the work can be extended. • Adopting Original Equipment Manufacturers (OEM) recommended maintenance – Applying OEM recommended maintenance can often yield quick returns in setting up in the CMMS. OEM maintenance will also need to be considered for new items under warranty management.

	• Re-grouping maintenance tasks to make planning easier and execution more effective and efficient – consider the way in which current maintenance tactics master data is grouped in the CMMS as there could be value gained though the re-structure of these items. For example, grouping multiple assets into one package of work for a zonal inspection.
1.8 Determine Operating Context	The MTD team is to be augmented by other Subject Matter Experts 's from multiple functions within the business as determined by the MTD Facilitator. The program of workshops is to be conducted in accordance with the project charter and plan. Project steps 1.8 – 1.14 inclusive shall be analysed within the workshop. The first task to be conducted in determining the operating context is to decide what the boundaries of the system are that are to be analysed. Process Flow, Piping, and Instrument Drawings can be used to assist in this process step. Before the maintenance tactics for an asset can be developed, a clear understanding of the operating context must be documented and clearly understood by the stakeholders participating in the development stage. Due consideration should be given to the aspects such as: • The primary purpose of the item and how it contributes to the business. • The criticality of the system, productive and maintainable assets. • The environmental conditions under which assets perform. • Whether the item is part of a batch or flow process • If redundancy exists. • Relevant performance and quality standards. • Safety standards applying to the asset. • Environmental standards. • Typical duty and operating cycles. • Typical maintenance and support arrangements. • Whether stockpiles exist which allow plant down time without affecting overall production. • Impact of repair time on production. The Operating Context statement should describe how the item operates at a high level and the analysis boundaries and interfaces with other systems. For example, a pump deployed within a single line operation process has a significantly different operating context to a similar item employed in the presence of redundancy. The example below provides further context.



Two different configured systems required to perform the same function. System 1 is a stand-alone system and needs to be operational in order to preserve system functionality. System 2 provides for a redundant item and therefore the operating context for that system is different, as it has built in redundancy. In order for each system to stop performing its function, in system one the stand-alone pump must fail, but in system two both pumps must fail and therefore the probability of system failure is significantly reduced. The maintenance strategy for each system will also differ as demonstrated below:



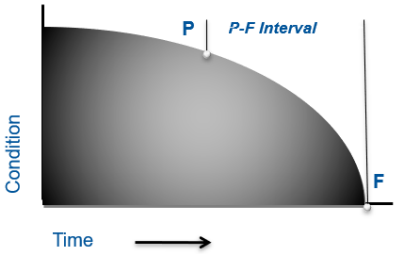
In System 1, failure would become evident under normal operating conditions and Condition Based Monitoring would be a highly recommended strategy to enable the detection of a failure before functional failure occurs.

In System 2, the duty pump failure would also be evident, but failure of the standby pump would be hidden, as the failure would not reveal itself until the onset of another failure, i.e., the duty pump. The maintenance strategy of the first pump could differ to the standalone pump in that it could be operated to failure, and analysis needs to be conducted to see if Condition Monitoring is cost effective. For the standby pump as failures are hidden the pump needs to be tested and fault-finding activity needs to

	occur in order to reveal any hidden failure and to ensure that the pump is capable of starting and performing as expected when required to do so.
1.9 Determine System Functional and Performance Requirements	In order to understand when a system has begun to or has entered into functional failure then knowledge of the functional and performance parameters is required for that system. Both Primary and Secondary functions shall be developed. For example, a conveying system Primary function is to transport product from A to B at a feed rate of x tonnes/hour and a secondary function is for the product to be contained within the system. By establishing the function and performance parameters then the analyst can easily decide which system and assets are important from a functional perspective. Primary functions should be listed first with the functions noted numerically.
1.10 Determine Functional Failures	Failed states are known as functional failures because they occur when an asset is unable to fulfil a function to a standard of performance which is acceptable to the user, which include partial functional failure, if the required system functionality cannot be performed. Functional Failures are analysed at two levels: • firstly, by identifying what circumstances amount to a failed state then, • secondly, by asking what events can cause the asset to get into a failed state. Typical Functional Failures of a hydraulic system would be: • *Unable to supply hydraulic oil to hydraulic systems at all.* • *Supplies less than 250 Litres per minute.* • *Supplies hydraulic oil at less than 125 bar.* • *Supplies hydraulic oil at greater than 135 bar* • *Unable to supply clean hydraulic oil to hydraulic systems.*
1.11 Determine Failure Modes	The development and use of Failure Mode models could be considered, where for example, models could be developed by asset classification, applied to actual working assets, and adjusted as necessary based on local conditions. Failure modes shall be identified for each functional failure identified at process step 1.9 and at a level of causation that makes it possible to identify an appropriate failure management policy. Each failure mode identified shall be credible in nature. Care must be taken to avoid focusing on ***symptoms*** when it is the ***cause*** of failure, which needs to be addressed. Consideration should be given for the use of formal methods for determining causation for high critical (A) items, for example structure tree analysis (cause and effect). The sources of likely failure modes for existing assets are: • Those, which have occurred before on the same asset. • Those currently being predicted or prevented by existing maintenance programmes. • Those that have not happened yet but are thought by the analysis team to be real possibilities in the context in question. Human factors shall also be considered when determining failure modes, such as:

	• Operating outside specifications • Poor quality workmanship Refer to Element E07 Operate for Reliability Manual, regarding the management of failures due to operating performance.
1.12 Determine Failure Effects	This should include all of the information needed to support the evaluation of the consequences of the failure, such as: • *evidence (if any) that the failure has occurred.* • *in what ways (if any) it poses a threat to safety or the environment* • *in what ways (if any) it affects production or operations* • *what physical damage (if any) is caused by the failure* • *what must be done to repair the failure.*
1.13 Determine Failure Consequences	Classified into four groups: • Hidden failure consequences: – no direct impact – expose the organisation to multiple failures with serious, often catastrophic, consequences. Generally associated to protective devices and will not be revealed until a secondary failure occurs. For example, gas detection device has failed and will not be revealed until a secondary failure occurs, gas leak, then a potentially fatal explosive atmosphere exists. • Safety and environmental consequences: – safety if could hurt or kill someone. – environmental if could lead to a breach of any corporate, regional, national, or international environmental standard. • Operational consequences: – affects production. – customer service – Includes operating costs (lost production) in addition to the direct cost of repair. • Non-operational consequences: Evident failures, which fall into this category, affect neither safety nor production, so they involve only the direct cost of repair. The level of significance of each failure is to be analysed and if the failure does have significant impact, then a preventive task should be carried out. Please refer to Appendix A RCM Decision Logic, this will aid in the determination of failure consequences and the strategy to be adopted to mitigate or significantly reduce the consequence.
1.14 Determine Maintenance Tactics	Selecting the most appropriate maintenance task is paramount in the RCM process and is highly dependent on the failure mode, failure effect and failure consequence and the ease at which functional failure can be detected. For each failure mode a maintenance type task shall be developed when technically feasible and worth doing (economically) RCM recognises various Maintenance Task types. • On-condition - entail checking for potential failures so that action can be taken to prevent a functional failure or to avoid its consequences. The extent to which this is possible is dependent on the P-F interval (the time between when the

potential failure, P, can be observed and the time at which the functional failure, F, will occur) as illustrated below:



The P-F interval is the interval between t
occurrence of a potential failure and its
decay into functional failure

On condition tasks must be done at intervals less than the P-F interval

- CBM is a valid approach when it is possible to monitor for a specific issue and establish a condition which may indicate potential failure.
- The P-F interval is reasonably consistent.
- It is practical to monitor the item at intervals less than the P-F interval.
- CBM Tasks should contain three elements:
  - The task instruction
  - The acceptable limits (quantified where practicable). Acceptable limit should be the limit that detects the potential failure and not functional failure. For example, vibration level on a bearing, the acceptable limit shall be set at a level where the potential point can be detected using appropriate monitoring equipment.
  - The action to be taken in the event of an unacceptable limit.

- Schedule Restoration - involves restoring the initial capability of an item or component at a specified age regardless of its apparent condition at the time. For restoration tasks to be technically feasible, the item or component must suffer from age-related failure (wear out) (e.g., a large electric motor, which needs to be rewound after a specified level of usage). See Appendix C for failure distribution patterns.
  - Schedule restoration tasks only contain the task instructions.

- Scheduled Discard - involves discarding an item or component at or before a specified age limit regardless of the condition at the time. For discard tasks to be technically feasible, the item or component must suffer from age-related (wear out) failures (e.g., wear plates may need discarding at a particular point in time, depending on the operating context). Where the associated loss of function could lead to a breach of a known environmental standard or regulation.
  - Schedule discard tasks only contain the task instructions.

- Fault Finding - tasks designed to check whether something still functions are known as Failure Finding Tasks (FFT).
  - Failure finding applies only to hidden or unrevealed failures. Hidden failures in turn generally only affect protective devices.
  - Scheduled failure-finding entails checking a hidden function at regular intervals to find out whether it has failed.
  - A hidden failure will not normally become apparent to the operations or maintenance staff in the normal course of operation.
  - Therefore a loss of function will *NOT be evident*

- A Hidden Failure only increases business exposure to the risk of a multiple failure happening concurrently.
- Cost and safety consequences of multiple failure should be considered when deciding if preventive maintenance tasks are cost effective
- The probability of a multiple failure in any given period is governed by:
  - probability that the protected function will fail
  - protecting device is in a non operational state at the time
- Probability of multiple failure can be reduced in one of four ways:
  - reduce the likelihood of failure of the *protected* device by carrying out effective *preventive* maintenance tasks.
  - increasing the availability of the *protecting* device by carrying out preventive maintenance tasks.
  - checking on a regular basis to see if the *protecting* device has failed
  - modify the asset to *eliminate* the hidden failure mode

- Operate to Fail (OTF) - In cases where the failure is evident and there is no appropriate scheduled task and the associated failure mode shall have no safety or environmental consequences. Where preventive or predictive tasks cannot be cost justified, meaning that over a period of time the cost of doing the task will be higher than the cost to the business of the consequences of the failure plus the repair cost. In these circumstance Operate to Failure strategy shall be selected.
  - OTF require no tasks, but consideration should be given to spares availability, lead time and returning the asset to service in a timely manner

- Re-design - is considered when proactive maintenance is not possible for failures which have safety or operational consequences, or proactive maintenance or failure finding is not possible for Hidden Functions and the failure would affect safety or the environment.

- Corrective Maintenance – are to be developed for on-condition tasks that are developed as part of the MTD process and when functional failure has occurred and found during fault finding tasks. It is recommended that both planned corrective and unplanned corrective tasks are developed due to the cost difference and modelled accordingly.

Again, the RCM logic diagram at Appendix A will provide assistance in this selection criteria.

A brief description of the maintenance task is to be developed bearing in mind that the purpose of the task development phase is to provide enough detail so that the end user is able to understand what needs to be done and not necessarily how to do it.

- Keep the task description as brief and concise as possible. If available picture, drawing etc. could be included to enhance understanding. When a high level of detail is required then simply refer to a separate Standard Work Procedure.
- Clearly identify the Productive Unit and Maintainable Item to which the task applies. Detail clearly, what must be done. For example, “Check the lubricating oil temperature” means little as it does not nominate what is acceptable. “Check lubricating oil temperature is not greater than 85° C and report immediately to shift supervisor if excessive temperature is detected” means far more.

- Use consistent terms and phrases in describing tasks and actions. For example, a "Test" may be defined as "To determine that a component or system is functioning correctly using appropriate test equipment". Establish an agreed glossary of terms before completing schedules.
- Be consistent in the structure you adopt for your schedules. Agree on a structure and stick to it.
- When writing your schedules use a level of detail consistent with the skill of the person carrying out the task. Do not attempt to teach a tradesman how to do their trade.

Consideration should be given to develop more prescriptive maintenance tasks if this will aid in the execution of the task. Each site must decide on the benefit of the development of prescriptive maintenance instructions in the form of Standard Work Procedure as mentioned above.

As part of the maintenance task determination, it is also important to understand under what conditions a corrective task is triggered. Corrective tasks are to be triggered before functional failure occurs, so the trigger points needs to be a point where there is enough time to plan, schedule and executed the work.

For example, if the failure point of a bearing from an over temperature failure mode was 100 degree C the corrective work should be triggered at some point in time before this functional failure point is reached. It is therefore important that some understand of P-F is know to assist with this determination.

Other items to include for each maintenace task include but not limited to:

- Uniques identifier for the schedule of work
- Task title that summarises the content and provides identification.
- Safety requirements
- Task Frequency
- Task Duratiuon
- Trade/Artisan Type
- System/Asset Online or Offline
- Specialst tools and equipment

It is a requirement that each maintenace task can be justified, the following provides informatation regarding typical requirements:

- Tasks meant to prevent Hidden Failures should reduce the risk of the multiple failure associated with that function to an acceptably low level. When a suitable preventive task cannot be found then a periodic fault-finding task should be performed.
- Tasks scheduled to prevent safety or environmental failures should reduce the risk of that failure to a very low level indeed. Redesign is mandatory when a task cannot be found to reduce the risk of failure to an acceptably low level.
- Tasks with operational consequences should cost less over a period of time than the cost of the operational consequences and the cost of repair over the same period. If there is not a cost-effective task, "Operate To failure" is the appropriate maintenance strategy.
- Failures with non-operational consequences must also be justified on economic grounds.

	• Resisting the temptation to carry out maintenance tasks just because there is a window of opportunity to do those tasks. For example, if there is a justifiable reason to adopt an OTF strategy then it should be taken.
1.15 Authorise Maintenance Tasks	Maintenance tasks are to be authorised by the respective manager.
1.16 Determine Sparing Requirements	All assets with maintenance task assigned will undergo a spares review. The primary aims will be to reduce the risk of stock-outs impacting on business performance and, on the other hand, to minimise the cost of holding unnecessary stock. Refer to Element E13 Procurement and Supply Chain Management Manual The review process will be conducted by maintenance and operations personnel together and will address: • How the maintenance requirements of the asset impact on spares requirements. A Level of Repair Analysis (LORA) should also be conducted to determine whether spares will be repaired or replaced and if so, by whom. • What the consequences are of not holding a spare • Whether the spares requirement can be anticipated ("dependent demand") • What stock holding is needed if the requirement for a spare cannot be anticipated ("independent demand"). Glencore Copper Critical Spares Standard shall be consulted to determine critical spares strategy.
1.17 Package Maintenance Tasks	The result of an MTD analysis is a set of recommendations for a series of discrete maintenance tasks. Task grouping is the process of combining those discrete tasks into an efficient, effective, and executable maintenance program. Concurrent tasks combined into a set of "work packages" The objective is to minimise downtime and optimise use of resources. A typical maintenance program is made up of Work Packages based on common recurring units of measure: • Calendar time • Operating units ○ operating hours ○ cycles ○ production units • "For convenience" • A Combination of the above Most CMMS systems have the capability to supress maintenance tasks (standard jobs), which provides for a hierarchical approach to maintenance (see example below), with the following advantages: • Ease of task scheduling where packages of work are contained within one maintenance plan. • Reduces the number of work orders from the CMMS, saving time and money in processing. Maintenance Hierarchy example: Tasks 1,2,3 are performed every 100 hours. Tasks 4,5 every 200 hours

	Tasks 6,7,8,9 every 400 hours Packages would look like this and contain only one work instruction per package: 100 hours: Tasks 1,2,3 200 hours: 1,2,3,4,5 (100-hour tasks suppressed) 300 hours: 1,2,3 400 hours: 1,2,3,4,5,6,7,8,9 (100 and 200 tasks suppressed) There is no definitive means by which tasks are packaged, the following will assist and influence on packaging requirements: • Task frequency – how often the task is to be performed. • Asset location – packed as systems or geographical locations? • Asset status – tasks executed with assets running or stopped. • Trades/skills – same trades/skills packaged together. • Tool and spares availability – lead-time could influence. • Planned downtime – for operational requirements for example. • Task duration – the time each package of work takes to complete. There are some rules that are mandated in the packaging process: • The interval of failure finding tasks or tasks, which manage safety consequences, will not be increased under any circumstances. • Each item to which the task applies will be fully identified. A typical example of a package of work: Maintainable Unit – Maintainable Function – Resource – System Status – Frequency 1234 - CV001 - MECH - ONLINE - 004W
1.18 Develop Work Instructions	A standard work instruction template should be developed giving consideration for some or all of the items below: • Related asset criticality rank (or priority) • Safety requirements (e.g., permit, PPE, and isolation requirements) • Access requirements • Access time • Related drawings and manuals • Whether the maintenance activity is to be conducted with the asset running or stopped • Total downtime required, if appropriate • Skills required to complete the Job (type and number) • Elapsed maintenance time to complete the Job. • Any specialised tools and equipment required (if any) • Materials and spares requirements (if any) • Failure modes under management • Steps needed to complete Job (e.g., “Remove guard” etc.).
1.19 Load Data into CMMS	The Master Data created during the RCM process shall be loaded/updated in the CMMS. Maintenance tasks shall be planned and scheduled according to Element 08 Maintenance Work Management Manual.
1.20 Analyse and Improve	Continuous improvement (optimisation) of maintenance tactics performance is a mandatory requirement of this Process Standard. This will be achieved in three ways:

- **Observe Asset Performance**

For High Criticality (A) assets, the functional failures and failure modes identified through the RCM analysis will be used as failure codes in the CMMS. This will assist maintenance staff in recording failure history.

- **Conduct Age Exploration, Failure Rate and Maintenance Cost Analyses**

Age Exploration: Age exploration is used to fill knowledge gaps where insufficient information was available on types and frequency of failure and age-frequency relationships at the time the RCM review was conducted. In such cases, age exploration will be undertaken to fill the knowledge gaps and further refine the RCM analysis.

- **Failure Rate Analysis:**

A failure rate analysis will be conducted to confirm initial assumptions and to seek opportunities for reliability improvement through redesign or more appropriate maintenance activities. Failure rate analysis will generally be the responsibility of Reliability Engineering.

- **Cost Analysis:**

Similarly, maintenance cost analysis reviews will be conducted to identify possible areas for cost reduction.

- **Formal Review**

A formal review of high critical (A) asset performance post-RCM will be conducted on an annual basis by the Facilitator. This will generally be actioned through the plant Maintenance Work Order system. Reports will be provided to the site Plant and Maintenance Managers.

The output from the RCM process provides input into other Glencore Copper Asset Management Framework Elements as follows:

- Element 03 Asset Management Planning – details regarding the maintenance strategy, activities and cost are to be captured in the Asset Management Planning process and contained within the Strategic Asset Management Plans.
- Element 08 Maintenance Works Management – the output from the RCM analysis provides an input into the CMMS in the form of master data for maintenance execution purposes.
- Element 04 Capital Investment and Decision Making – the output from the RCM analysis provides indicative cost and timing for corrective work, of which some would be capital investment for replacement assets.
- Element 09 Shutdown and Overhaul Management – output from the RCM analysis provides a list of tasks that are to be executed within the Shutdown and Overhaul process.
- Element 13 Supply and Procurement – the output from the RCM analysis provides for a list of recommended spares that are to be managed according to Element 13.
- Element 14 Risk Management and Compliance – RCM being a risk-based approach process has a natural tie in with E14.

4.3 Process RACI Description

The RACI matrix below in Table 4-2 addresses the high-level functional elements of the Maintenance Tactics Development process. For the overall Asset Management RACI matrix, refer to the Asset Management Intranet page.

Site roles may differ to the roles identified in this RACI, these roles however should be used as a guide for the development of a site-specific RACI, while maintaining the intent and content of the Manual.

Note: The HSE and Commercial Departments do not appear in this high-level RACI, although their involvement will be required from time to time dependent on the nature of the variance.

The codes used in the RACI analysis are defined as follows:

TABLE 4-2 PROCESS RACI

R	Responsible	The "doer" – the individual(s) who will perform an activity or directs others to perform the activity.
A	Accountable	"The buck stops here" –the individual who is ultimately "owns" the activity – there should only be one person
C	Consulted	person that needs to feedback and contribute to the activity
I	Inform	The individual(s) who needs to be informed after a decision or action is taken if relevant. (Via 1-way communication)

Process Step \ Functional Role	Reliability/ Maintenance Management	Operations	Maintenance Execution	Reliability /Maintenance Engineering	Maintenance Planning	Technical Services	Supply
1.1 Identify Systems/s to be Analyses	A			R	C		
1.2 Project Set Up	A			R			
1.3 Get Technical Hierarchy from CMMS	A			C	R		
1.4 Load Technical Hierarchy into RCM Tool	A			R	C		
1.5 Gather and Analyse Performance, Downtime and Work Order Data	A	C	C	R	C	C	C
1.6 Assemble MTD Team	A	C	C	R	C	C	C
1.6.1 Is the item Criticality High	A			R			
1.6.2 Is the Item New	A			R			
1.6.3 Is the Item Criticality Medium	A			R			
1.7 PMO	A	C	C	R	C	C	
1.8 Determine Operating Context	A	C	C	R	C	C	
1.9 Determine Asset Function and Performance Standard	A	C	C	R	C	C	
1.10 Determine Functional Failures	A	C	C	R	C	C	
1.11 Determine Failure Modes	A	C	C	R	C	C	
1.12 Determine Failure Effect	A	C	C	R	C	C	
1.13 Determine Failure Consequences	A	C	C	R	C	C	

Process Step / Functional Role	Reliability/ Maintenance Management	Operations	Maintenance Execution	Reliability /Maintenance Engineering	Maintenance Planning	Technical Services	Supply
1.14 Determine Maintenance Tasks	A	C	C	R	C	C	
1.15 Authorise Maintenance Tactics	A			R			
1.16 Determine Spares Requirements	A		C	R	C	C	C
1.17 Package Maintenance Tasks	A	C	C	R	C	C	
1.18 Develop Work Instructions	A	I	I	R	C	I	
1.19 Load Data to CMMS	A			C	R		
1.20 Analyse and Improve	A	C	C	C	R	C	C

TABLE 4-2 FUNCTIONAL ROLE DESCRIPTIONS

Function	Description
Reliability/Maintenance Management	Provides leadership in the short to medium term time horizon. Accountable for establishing the maintenance plan, budget and managing the execution of maintenance activities. Ensures that the process for the development of MTD is conducted in accordance with this procedure and that the results are entered into the CMMS.
Operations	Provides advice and information regarding the analysis of MTD and is a working member of the MTD team.
Maintenance Execution	Provides advice and information regarding the analysis of MTD and is a working member of the MTD team.
Reliability/Maintenance Engineering	Responsible for the development and implementation of MTD Maintenance Strategies at the local level. Is the Facilitator of the Workshop/s. Provides Coaching and Mentoring to team members on the MTD process. Is responsible for the update and authorisation of changes to maintenance tactics.
Maintenance Planning	Provides advice and information regarding the development and implementation of MTD maintenance strategies and is a working member of the MTD team. Responsible for extracting and loading master data to the CMMS.
Technical Services – Metallurgists, Process Engineers and Mining	Provides advice and information regarding the analysis of MTD maintenance strategies and is a working member of the Criticality Workshop team. Provides expert advice regarding production processes to support functions and performance standard development.
Supply	Provides expert advice regarding the selection of spare parts for MTD maintenance strategy development.

5. Process Benefits

Some of the notable benefits of a reliability-based approach to maintenance include:

- A better understanding of the required functions and performance standards of the systems within the operating business.
- Minimisation of failures that have adverse business consequences.
- Improved asset reliability and availability for critical assets.
- Improved facility productivity and operating performance for critical systems.
- Fewer failures caused by unnecessary maintenance activities.
- Greater safety and environmental awareness.
- Clear strategies for preventing Failure Modes that can affect safety.
- Longer useful life of expensive assets.
- Greater maintenance cost-effectiveness.
- More emphasis on the maintenance requirements of critical items.
- Better knowledge level of systems and assets.
- Reduced reliance on outside experts because staff have a better understanding of the plant.
- Better teamwork and communication between departments.
- Documented history of the logic taken in assigning maintenance tasks and intervals.

6. Deployment

The deployment of the Maintenance Tactics Development approach across all business areas is through:

- Awareness presentation
- Position description alignment
- Competency assessment
- Training, coaching, and mentoring

6.1 Awareness Presentation

This Procedure is supported by a slide presentation and video to help facilitate communication and to raise awareness on how it contributes to Maintenance Tactics Development and business success.

6.2 Position Description Alignment

To effectively deploy this standard, the essential process steps and activities need to be diligently mapped and assigned to approved positions within Glencore Copper.

The RACI matrix assists in the visualisation that everyone is working towards the same goal. It clarifies not only the individual's responsibilities, but also that of others. It has proven to be an effective tool to quickly understand and close the gaps between current and desired practices and business process activities.

Position descriptions including KPIs need to be aligned to the RACI matrix and measures of success. This is the basis for ongoing performance reviews and the creation and execution of individual development plans.

6.3 Competency Assessment

This standard is supported by an Asset Management competencies assessment process and each site is responsible for assessing Asset Management competencies:

- Guide recruitment, selection, development, and deployment of people in asset management roles across the company
- Provide employees responsible for managing assets with feedback, development, coaching and recognition for their performance.
- Map, build and manage the asset management talent pool across the company.
- Identify strengths and development needs in asset management practitioners.

6.4 Training, Coaching and Mentoring

Coaching and mentoring is another way to assist employees improve their understanding of their role and to improve performance.

Coaching is generally about improving performance in the short term where mentoring is about achieving high performance and career aspirations over a longer term.

Coaching or mentoring requires addressing that a person is not currently up to the required level to perform their role as effectively and efficiently as they could. The build-up of skills to perform efficiently and effectively requires a large investment of time and cooperation.

Coaches and Mentors need to be carefully selected and must be committed for the long term to effect maximum success.

7. Improvement

The process to review and continually improve the approach and deployment for Maintenance Tactics Development is scheduled annually or when there is a sudden change in the Business and Asset Management objectives. As the organisation matures a longer review period could be considered.

8. Maintenance Tactics Development Review

Maintenance tactics for High Critical (A) assets should be reviewed annually or when significant changes occur in the business that could prompt a change in maintenance tactics, for example, chronic system or assets failure resulting in significant business impact event.

9. References

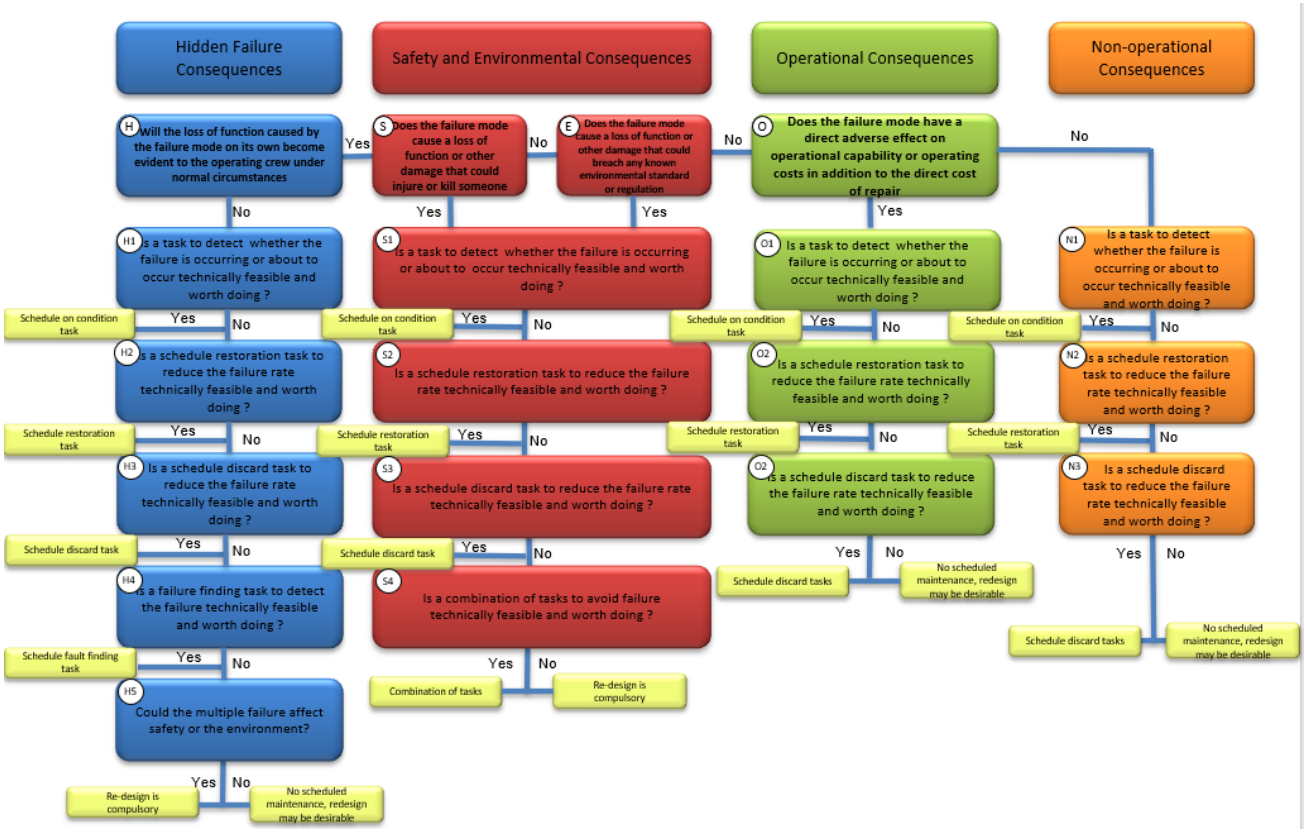
The following documents provide detailed support material that should be read in conjunction with this standard to provide a comprehensive understanding of Maintenance Tactics Development.

DOCUMENT TITLE	DESCRIPTION
Asset Management Framework	This document defines the overall direction of asset management across Glencore Copper.
Equipment Operating and Maintenance Strategy Manual Element 05	This document defines the process for the development and implementation of Equipment Operating and Maintenance Strategies within Glencore Copper
Glencore Copper Corporate Risk Matrix	The multi-criteria risk matrix that is used to determine corporate risk through consequence and likelihood evaluation of potential failure events.
SAE JA 1011: Evaluation Criteria for Reliability-Centred Maintenance Processes.	Developed by the commercial aviation industry to improve the safety and reliability of their assets. Because of this, it's known as a Reliability-Centred Maintenance (RCM) process.
RCM II: John Mowbray	Considered to be one of the leading literary works on Reliability Centred Maintenance.

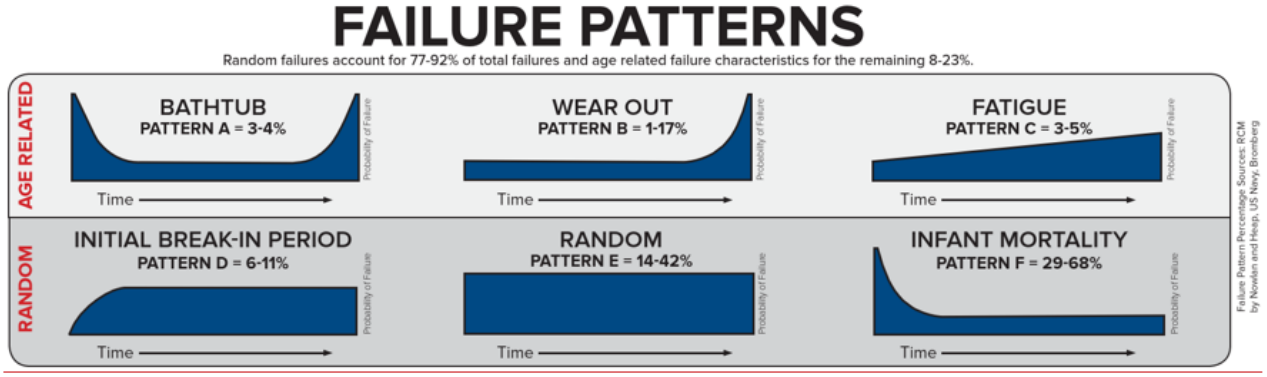
10. Revision History

VERSION NO.	REASON FOR REVISION	DATE
0.1	New Standard to support Asset Management Manual	25/03/2021
0.2	Gather Documents and Create Draft Procedure	04/08/2021
0.3	Review and Signoff 1st Draft Procedure with Sites	13/08/2021
0.4	Reformat and Issue for Final Review	14/09/2021
1.0	Update Site Signoff and Publish	07/10/2021

11. Appendix A – RCM Decision Logic



12. Appendix B – Failure Distribution Patterns



13. Appendix C – Stakeholder Sign-Off

Stakeholder Name	Site or Department	Contact	Stakeholder Signoff
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Maintenance Tactics Development Standard

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